

Distributed Inclusion Dependency Discovery

Jovial Vernon Dsilva, Bindu Sampath

Philipps-Universität Marburg

Architecture

- 1. Guardian** – The root actor that manages the entire cluster lifecycle and coordinates graceful shutdown across all nodes
- 2. DependencyMiner** – The central coordinator that generates IND candidates, distributes validation tasks to workers, and collects results. Acts purely as a coordinator without storing any column data
- 3. InputReader(s)** – Reads CSV files in batches and streams data to the master for distribution to workers
- 4. DependencyWorker(s)** – Stores only assigned columns using vertical partitioning and validates IND candidates through conservative subset checking
- 5. LargeMessageProxy** – Handles messages exceeding 256KB network limit by chunking and reassembly, enabling distributed execution
- 6. ResultCollector** – Receives validated INDs from DependencyMiner and writes them to the output file

Data Flow & Design Decisions

Data Flow:

- **LOAD** - InputReaders stream CSV data in batches to the master coordinator
- **PARTITION** - Master forwards batches to all workers via LargeMessageProxy, each worker extracts and stores only its assigned columns using round-robin distribution
- **PRUNE** - DependencyMiner applies semantic pruning by generating candidates only from KEY columns
- **VALIDATE** - DependencyWorkers perform conservative validation using subset checking, rejecting candidates when both required columns are not owned locally
- **COLLECT** - Valid inclusion dependencies are sent to ResultCollector and written to the output file

Key Design Decisions:

- Conservative validation preserves strict partitioning by rejecting INDs when both columns aren't owned locally, ensuring architectural correctness at the cost of potentially missing some valid dependencies
- Batch broadcasting allows the master to forward data to all workers while workers only persist their assigned columns, maintaining true vertical partitioning without centralized storage
- LargeMessageProxy enables distributed execution by handling network message size constraints through automatic chunking and reassembly
- Round-robin column assignment ensures even distribution of columns across workers for balanced workload
- Semantic pruning focuses validation on KEY columns only, reducing the search space while targeting meaningful foreign key relationships